Miriam Muthoni Muthii

Full-Stack Software Engineer

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Summary

A passionate Full-Stack Software Engineer who enjoys building complete, functional applications from the ground up across diverse domains and use cases. Brings hands-on experience working across the full stack using JavaScript, TypeScript, Angular, React, and Next.js on the frontend, and Python and Java on the backend, delivering production-quality solutions that balance performance, reliability, and clean code. Thrives in collaborative, remote environments and takes genuine satisfaction in seeing software go from an idea to something real that people actually use.

Core Competencies

TypeScript & JavaScript	TypeScript architecture, Next.js (SSR, API routes), React (TypeScript), Angular (TypeScript), Vue.js, JavaScript, component-based design, state management
Backend, APIs & Databases	Python (FastAPI, DRF), Java, Spring Boot, RESTful APIs, PostgreSQL, MySQL, SQL schema design, query optimisation, SQLAlchemy ORM, microservices
Caching & Real-Time Data	Redis for caching, session management, and real-time data handling; efficient data flow between frontend, caching layer, and database
Code Quality & Practices	Clean code principles, maintainable and well-documented codebases, code review, best practices adherence, test-driven development
AI Integration	LLM integration into TypeScript workflows, prompt engineering, agentic pipelines, LLM output evaluation, CrewAI, OpenAI APIs
Remote Collaboration	Distributed team environments, async communication, cross-functional coordination, clear technical documentation and verbal updates
DevOps & Infrastructure	Docker, Podman, CI/CD (Azure DevOps), Azure Container Registry, Git
Observability & Monitoring	OpenTelemetry for instrumentation and tracing, Prometheus for metrics collection, Grafana for visualization and monitoring dashboards
UI/UX Design	Figma, Miro, Adobe XD; user journey mapping, wireframing, prototyping
Client & Stakeholder Engagement	Requirements translation, solution presentations, technical consulting, cross-functional collaboration

Experience

Software & AI Developer *Aug 2025 – Present*
CCI Global *Nairobi, Kenya*

- Designed and built complete, production-quality full-stack applications from scratch, implementing end-to-end functionality including data management, analytics pipelines, real-time features, and automated reporting across diverse use cases.
- Architected scalable TypeScript-driven frontend systems using Angular, integrated with Python backend services via RESTful APIs, ensuring efficient and reliable data flow across the full application stack.
- Designed and implemented SQL database schemas using SQLAlchemy ORM, consolidating sessions, evaluations, and evidence into a centralised data store, reducing manual reporting by 50% and enabling real-time data access.
- Built dynamic dashboards and data-driven platforms demonstrating varied architectural patterns, from component-based frontend design through to backend service integration and database optimisation.
- Integrated LLM workflows and agentic AI pipelines into the TypeScript codebase, implementing real-time data handling and automated evaluation features across production conversations.
- Participated in code reviews, ensuring adherence to clean code principles and coding standards, fostering a culture of quality and continuous improvement across the engineering team.
- Collaborated in a cross-functional remote team environment, maintaining clear technical documentation and providing regular progress updates to stakeholders.

Software Developer, UI/UX Focused *Feb 2025 – Jun 2025*
NextPro Business Solutions *Nairobi, Kenya*

- Designed end-to-end USSD user journeys and interaction flows using Miro for workflow mapping and Figma for UI/UX prototyping, translating business requirements into development-ready specifications.

- Developed functional specifications, screen flow documentation, and structured technical documentation to align stakeholders with development teams.
- Implemented validation logic enforcing business rules, improving data integrity and system reliability.
- Coordinated with clients and cross-functional teams, managing expectations and ensuring smooth delivery throughout the project lifecycle.

Software Developer *Jan 2024 – Dec 2024*
E&M Technology House *Nairobi, Kenya*

- Designed and delivered complete, production-ready SaaS applications from scratch using Next.js, React, and Angular, implementing diverse features including user authentication, data management, business logic, and reporting across multiple client domains.
- Architected full-stack systems with TypeScript frontends, Java Spring Boot backends, and PostgreSQL databases, applying SQL schema design, query optimisation, and efficient data flow patterns throughout.
- Built varied application types including management systems, workflow tools, and data-driven platforms, each demonstrating different architectural patterns and feature implementations.
- Mentored junior developers on TypeScript and full-stack best practices, conducted code reviews, and oversaw project delivery as a technical mentor and lead.
- Maintained CI/CD practices and Git-based version control across concurrent projects in Agile, distributed team environments.

Relationship Officer, Operations *May 2021 – Dec 2023*
Equity Bank *Nairobi, Kenya*

- Managed high-volume corporate transactions with 98% drawer balance accuracy under strict compliance controls.
- Built and managed client relationships, resolving escalations and maintaining high service levels across individual and corporate accounts.

Education

BSc. Biochemistry, First Class Honours *Sept 2017 – Dec 2021*
University of Nairobi *Nairobi, Kenya*

Skills & Interests

- **Full-stack application builder:** Experienced designing and building complete, production-quality applications from scratch across diverse domains, implementing end-to-end functionality from database schema through to frontend delivery.
- **Modern frontend development:** Experienced building production systems across React, Angular, Next.js, and Vue.js, with a strong instinct for scalable architecture, server-side rendering, and clean, maintainable code.
- **AI integration on applications:** Experienced integrating LLM workflows, agentic pipelines, and prompt engineering directly into production applications, adding intelligent capabilities to full-stack systems.
- **Remote-first mindset:** Experienced working in distributed, cross-functional team environments with a strong emphasis on written communication, documentation, and async collaboration.
- **Database and caching expertise:** Solid experience with SQL schema design, PostgreSQL, and Redis for caching and real-time data handling, ensuring performant and reliable data flow across full-stack systems.
- **Performance and reliability focus:** Actively optimises frontend and backend modules for performance, reliability, and scalability as a core part of the development process.
- **Continuous improvement:** Committed to growing technical depth, staying current with the broader software engineering ecosystem, and contributing to a culture of learning within teams.

References

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